# Week 6: GANs

Teaching Neural Networks to Create

**Live Coding Session — Week 6**

# The Big Shift: Classify → Create

*Weeks 1–5 taught models to recognize. Week 6 teaches them to imagine.*

**Discriminative (Weeks 1–5)**

Input → Label

Learns P(y | x) — "what class is this?"

Examples: MLP, CNN, Transfer Learning

*Classifies existing data*

**Decision boundary**

→

**Generative (Today!)**

Noise → New Data

Learns P(x) — "what does this data look like?"

Examples: GAN, VAE, Diffusion Models

*Creates new data*

**Data distribution**

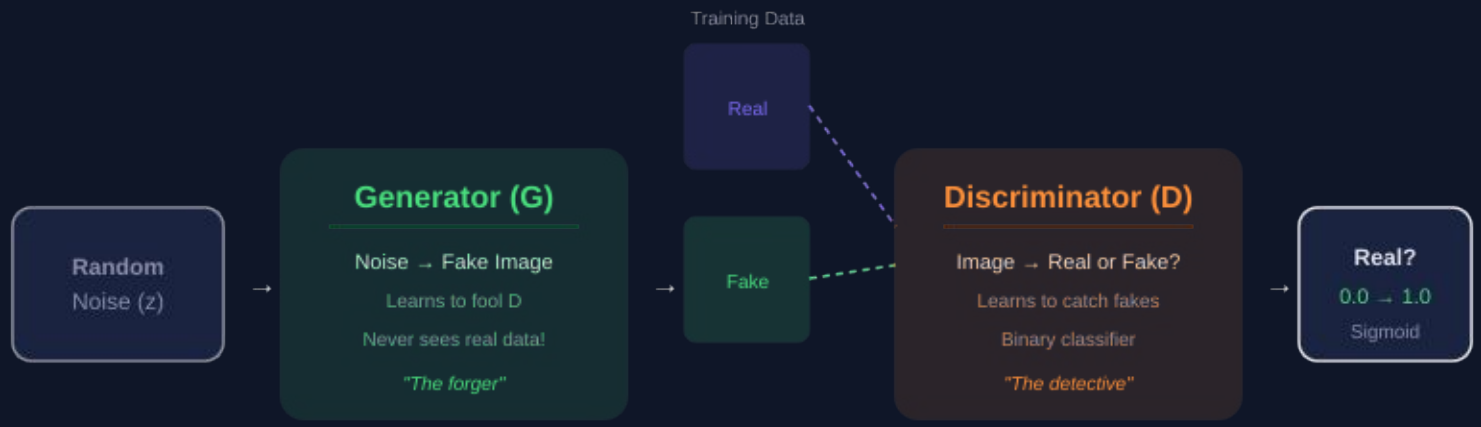
**How do you train a model to create?** You can't just compute MSE against a target — there's no single "right" output.

**GAN solution:** Train a second network (Discriminator) to judge quality. The Generator learns from the Discriminator's feedback — not from the data directly.

***The Generator has never seen a real image.*** *It only learns from the Discriminator's feedback.*

# The GAN Architecture

*Two networks competing: a forger and a detective*




**The Adversarial Game**

G tries to maximize D's mistakes. D tries to minimize its mistakes. They push each other to improve.

**Both are just nn.Module networks.** Same Linear/ReLU/Sigmoid layers from weeks 1–2. Same Adam optimizer from week 5. Nothing new — just a new training game.

# The Training Loop: Two 5-Step Loops

*Same forward → loss → zero_grad → backward → step. Just done twice per batch.*

### Step 1: Train Discriminator

1. Feed real images → D should output 1
2. Generate fake images → D should output 0
3. Compute loss on both
4. Backprop through D only

```
real_loss = BCE(D(real), ones)
fake_loss = BCE(D(G(z)), zeros)
d_loss = real_loss + fake_loss
d_loss.backward() → d_optimizer.step()
```

→

### Step 2: Train Generator

1. Generate fake images
2. Feed to D → G wants D to output 1
3. Compute loss (fool D)
4. Backprop through G only

```
fake = G(z)
g_loss = BCE(D(fake), ones)
← "pretend fakes are real"
g_loss.backward() → g_optimizer.step()
```

**Key insight:** When training G, we pass ones as labels — we're telling the loss function "these fakes are real." G then gets gradients that push it toward making more realistic images.

**D and G have separate optimizers.** We alternate: one step for D, one step for G. They improve together.

# Today's Coding Session

~15 min

## MLP GAN

Build G and D from scratch. Train on MNIST. Watch digits emerge from noise.

~15 min

## DCGAN

Conv-based GAN. Same game but with spatial features. Sharper results.

~5 min

## Epoch Progression

Side-by-side: MLP vs DCGAN quality. Epoch-by-epoch improvement grid.

**+ 10 min Q&A at the end**

# Questions?

Week 6: Generative Adversarial Networks

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